

CONCEPT PAPER

Development of a Mobile-Based Medication Adherence and Chronic Disease Monitoring System

1. Introduction

The advancement of information and communication technology has greatly transformed the healthcare sector by improving patient management, communication, and access to medical services. One of the major challenges faced by patients suffering from chronic diseases such as Diabetes, hypertension, asthma, and heart diseases is failure to adhere to prescribed medication schedules and medical appointments.

Medication non-adherence can lead to worsening health conditions, increased hospital admissions, complications, and even death. Many patients forget to take their medication on time, fail to attend medical reviews, or lack proper guidance regarding drug usage and treatment procedures.

This project therefore proposes the development of a Mobile-Based Medication Adherence and Chronic Disease Monitoring System that will help patients follow medication schedules, receive medical reminders, and improve communication between healthcare providers and patients through secure notifications.

The system is intended to primarily support patients with Diabetes but can also be extended to other chronic illnesses.

2. Background of the Study

Chronic diseases require continuous monitoring, regular medication, and periodic medical consultations. However, traditional methods of patient follow-up heavily rely on physical appointments and manual communication, which may not effectively support continuous care.

Many patients miss medication doses because they forget the time, misunderstand prescriptions, or fail to prepare necessary requirements such as meals or water before taking medicine. Similarly, healthcare workers face challenges in monitoring patient progress and reminding patients about consultations and checkups.

Mobile health systems have become important tools in improving healthcare delivery by enabling real-time reminders, automated notifications, and digital patient monitoring. The proposed system seeks to provide an intelligent and centralized platform that enhances medication adherence and hospital-patient coordination.

3. Problem Statement

Patients suffering from chronic illnesses face several challenges in adhering to treatment plans and attending scheduled medical consultations. The major problems include:

- Forgetting to take medication at the correct time
- Missing hospital appointments and medical reviews
- Poor understanding of medication instructions and prescriptions
- Limited monitoring of patients outside the hospital environment
- Lack of timely reminders from healthcare providers
- Poor coordination between hospitals, doctors, and patients
- Increased risk of complications due to non-adherence to treatment

These challenges negatively affect patient health outcomes and increase healthcare costs for both patients and medical institutions.

4. Aim of the Project

To develop a mobile-based Medication Adherence and Chronic Disease Monitoring System that assists patients in following medication schedules, attending consultations, and improving communication between healthcare providers and patients.

5. Objectives of the Project

General Objective

To design and implement a medication adherence system for improving treatment compliance and monitoring of patients with chronic diseases.

Specific Objectives

The project aims to:

1. Develop a system that reminds patients when to take medication.
2. Provide medication instructions and descriptions for each prescribed drug.
3. Notify patients in advance before medication time to allow preparation.
4. Enable hospitals and doctors to monitor patient treatment schedules.
5. Automate reminders for hospital consultations and medical checkups.
6. Create a secure hierarchical management structure for hospitals, administrators, supervisors, doctors, and patients.
7. Improve communication between healthcare providers and patients through notification-based alerts instead of direct messaging.
8. Enhance adherence to treatment plans among chronic disease patients.

6. Proposed Solution

The proposed solution is a mobile and web-based healthcare management system designed to support medication adherence and chronic disease monitoring.

The system will operate under different levels of users and responsibilities as described below:

6.1 Super Admin (System Owner)

The Super Admin will manage the entire platform and control all operations within the system. Responsibilities include:

- Registering and managing hospitals
- Monitoring overall system activities
- Managing system security and access
- Controlling hospital enrollment into the platform

6.2 Hospital Administrator

Each enrolled hospital will have its own system administrator responsible for:

- Managing hospital information
- Registering supervisors and doctors
- Monitoring hospital activities within the system
- Managing patient enrollment records

6.3 Supervisor Doctors

Supervisor doctors will:

- Manage and oversee assigned doctors
- Assign doctors to patients
- Monitor patient treatment progress
- Review adherence reports and notifications

6.4 Doctors

Doctors will:

- Enter patient prescriptions and medication schedules
- Set reminders for medication intake
- Schedule consultation dates and medical reviews
- Send medical notifications to patients
- Monitor patient adherence reports

Communication between doctors and patients will remain professional and controlled through structured notifications rather than open direct messaging.

6.5 Patients

Patients will use the mobile application to:

- Receive medication reminders
- View prescription instructions and dosage information
- Receive advance alerts before medication time
- Receive reminders for consultations and hospital visits
- Monitor their medication schedules

- Confirm medication intake where necessary

For example, the application may notify a patient:

“You are expected to take your medication in the next two hours. Please prepare drinking water and follow the prescribed dosage instructions.”

The system will also display information about the medication type, dosage instructions, and precautions.

7. Key Features of the System

The proposed system will include the following features:

- User authentication and role-based access
- Hospital registration and management
- Doctor and supervisor assignment
- Patient registration and management
- Medication scheduling and reminders
- Drug description and instruction display
- Consultation and checkup reminders
- Notification-based communication
- Patient adherence tracking
- Report generation and monitoring dashboard

8. Significance of the Study

Patients

- Improved adherence to medication schedules
- Reduced chances of missing treatment
- Better health monitoring and management
- Timely reminders for consultations and checkups

Hospitals and Healthcare Providers

- Easier patient monitoring
- Improved coordination between healthcare workers
- Better management of chronic disease patients
- Reduced workload through automation

Doctors

- Efficient follow-up of patient treatment progress
- Ability to monitor medication adherence remotely
- Improved communication with patients through notifications

Researchers and Developers

- Provides practical experience in mobile and healthcare system development
- Serves as a reference for future healthcare technology projects

9. Scope of the System

Functional Scope

The system will cover:

- Medication reminder management
- Patient registration and monitoring
- Hospital enrollment and management
- Doctor and supervisor management
- Consultation scheduling and reminders
- Notification management
- Prescription and dosage management

Geographical Scope

The system can initially be implemented in selected hospitals and later expanded to multiple healthcare institutions.

Content Scope

The project focuses on chronic disease management, particularly Diabetes, but can support other chronic illnesses.

10. Methodology

The project will follow the System Development Life Cycle (SDLC), which includes:

1. Requirement gathering and analysis
2. System design
3. Database design
4. Mobile and web application development

5. Testing and debugging

6. System implementation

7. Maintenance and evaluation

Data collection methods may include interviews, observation, questionnaires, and literature review.

11. System Requirements

Hardware Requirements

- Smartphone for patients
- Computers for hospital administrators and doctors
- Reliable internet connection
- Server or cloud hosting infrastructure

Software Requirements

- Mobile Application Framework (Flutter or React Native)
- Backend Framework (Laravel, Node.js, or Django)
- Database Management System (MySQL or PostgreSQL)
- Web Browser
- Notification services (SMS or Push Notifications)

12. Feasibility Study

Technical Feasibility

The required technologies for mobile application development, databases, and notifications are available and widely used.

Economic Feasibility

The project is cost-effective because open-source tools and frameworks can be used during development and deployment.

Operational Feasibility

The system will be easy to use for patients, doctors, and administrators through a simple and user-friendly interface.

Schedule Feasibility

The project can reasonably be completed within a period of approximately 10–14 weeks depending on system complexity.

13. Expected Outcomes

At the completion of the project, the following outcomes are expected:

- A functional medication adherence mobile application
- Improved treatment compliance among chronic disease patients
- Reduced missed medication doses and appointments
- Better patient monitoring by healthcare providers
- Improved healthcare service delivery through automation

14. Conclusion

The proposed Medication Adherence and Chronic Disease Monitoring System will provide an effective technological solution for improving treatment compliance and

healthcare monitoring. By automating medication reminders, consultation scheduling, and patient monitoring, the system will enhance healthcare delivery and improve the quality of life for patients suffering from chronic diseases such as Diabetes.

The project is technically, operationally, and economically feasible and has the potential to significantly contribute to modern digital healthcare systems.